Bayesian Modeling and Online Learning of Personalized and Time-Varying Treatment Effects in Non-Stationary Mobile Health Data

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Outline

Intensive Longitudinal Data and the ARC Study

- Smartphones and wearables enable **repeated, fine-grained** measurement of mental health in everyday life

Intensive Longitudinal Data and the ARC Study

- Smartphones and wearables enable fine-grained measures: **ILD**
- **Advancing Resilience and Coping (ARC) study**: a 4-week mobile health study for individualized suicide prevention
  - Ecological Momentary Assessment (EMA) surveys multiple times per day: suicidal ideation, affect, coping behavior
  - Passive sensing: GPS, phone usage, Fitbit activity & sleep

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Statistical Challenges

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Statistical Challenges

1. Within-person **temporal dependence**

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Statistical Challenges

1. Within-person **temporal dependence**
2. Between-person **heterogeneity** in risk, response, and dynamics

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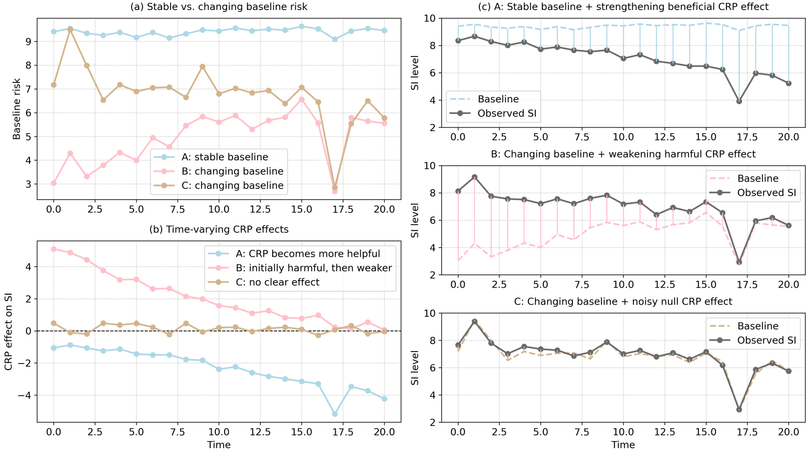
Statistical Challenges

1. Within-person **temporal dependence**
2. Between-person **heterogeneity**
3. **Time-varying treatment effects**
4. **Latent subgroups** (who benefits? who doesn't? Stable?)

The Challenge: Distinct Mechanisms, Similar Observations



Key point: Superficially similar observed trajectories may arise from distinct underlying mechanisms, requiring different intervention strategies.

Existing Approaches and Their Limitations

Approach	Strength	Limitation
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State-space models	latent dynamic processes; nonstationary trajectories	fitted separately: no pooling; fitted jointly: oversmooth important differences
Hierarchical Bayesian models	information sharing; uncertainty quantification	static subject-specific parameters

Each of them addresses part of the problem, but none combines all the features we need.

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No existing framework jointly integrates **nonstationary latent state evolution**, **individualized time-varying treatment effects**, **latent subgroup discovery**, and **principled information sharing** across individuals.

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Fundamental Tradeoff

Population-level stability $\longleftrightarrow$ Individual-level flexibility

Our goal: balance these via a *hierarchical Bayesian state-space model*

Proposed Model: Observation Equation

For individual i at time t:

$$\boxed{y_{i,t} = \underbrace{\alpha_{i,t}}_{\text{baseline risk}} + \underbrace{\beta_{i,t}\, x_{i,t}}_{\text{time-varying effect}\times\text{treatment}} + \underbrace{\boldsymbol{z}'_{i,t}\boldsymbol{\eta}}_{\text{covariates}} + \epsilon_{i,t}}$$

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- $y_{i,t}$: observed outcome (suicidal ideation severity)
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- $\epsilon_{i,t} \sim \mathrm{N}(0, \sigma_y^2)$: observation noise

The observed outcome is a **noisy combination** of the latent risk process, treatment response, and contextual factors.

Proposed Model: State Evolution

The latent states evolve as **individual-specific random walks**:

$$\boxed{\boldsymbol{\theta}_{i,t} = \begin{pmatrix} \alpha_{i,t} \\ \beta_{i,t} \end{pmatrix} = \begin{pmatrix} \alpha_{i,t-1} \\ \beta_{i,t-1} \end{pmatrix} + \boldsymbol{\omega}_{i,t}, \qquad \boldsymbol{\omega}_{i,t} \sim \mathrm{N}\left(\mathbf{0}, \begin{pmatrix} \sigma^2_{\alpha,i} & 0 \\ 0 & \sigma^2_{\beta,i} \end{pmatrix}\right)}$$

- $\sigma^2_{\alpha,i}$ controls how quickly baseline risk changes for person i
- $\sigma^2_{\beta,i}$ controls how quickly treatment effect changes for person i

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The model *adaptively learns* whether each person's trajectory is static or dynamic.

Subject-Level Classification Structure

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- G_i characterizes the **initial treatment effect direction**

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- ($D_{\alpha,i}$, $D_{\beta,i}$) characterize the **temporal dynamics**

$$\sigma^2_{\alpha,i} \mid D_{\alpha,i} \sim \underbrace{(1 - D_{\alpha,i})\delta_0}_{\text{static}} + \underbrace{D_{\alpha,i}\mathrm{Exp}(\lambda_\alpha)}_{\text{time-varying}}$$

$$\sigma^2_{\beta,i} \mid D_{\beta,i} \sim \underbrace{(1 - D_{\beta,i})\delta_0}_{\text{static}} + \underbrace{D_{\beta,i}\mathrm{Exp}(\lambda_\beta)}_{\text{time-varying}}$$

Subject-Level Classification Structure

A 3×2 subgroup structure focusing on $(G_i,\ D_{\beta,i})$ captures clinically meaningful treatment-response patterns:

Table 1: Subject-level subgroup structure and clinical interpretation.

G_i	$D_{\beta,i}$	**Interpretation**	**Clinical Implication**	**Prior mean prob.**
0	0	No effect (static)	Consider alternative strategies	0.25
0	1	No effect (dynamic)	Monitor closely before changing	0.25
-1	0	Beneficial (static)	Continue treatment	0.125
-1	1	Beneficial (dynamic)	Continue with close monitoring	0.125
$+1$	0	Harmful (static)	Consider discontinuation	0.125
$+1$	1	Harmful (dynamic)	Monitor and adjust	0.125

- $G_i \in \{-1, 0, +1\}$: direction of initial treatment effect
- $D_{\beta,i} \in \{0, 1\}$: whether treatment effect evolves over time
- Subgroup membership is **learned from the data**

Graphical Model Overview

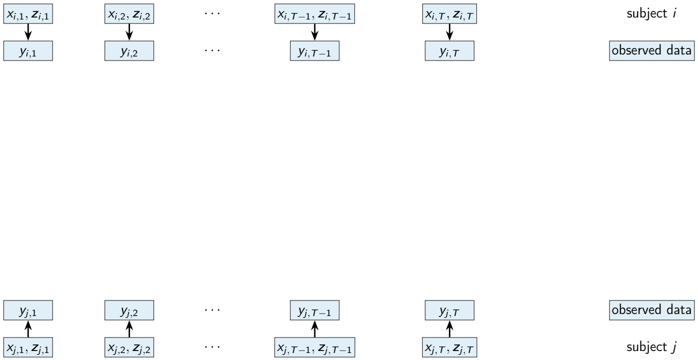

Figure 1: Hierarchical Bayesian state-space model: observed data are linked to time-varying latent states within each individual, while a higher-level classification structure groups individuals into clinically meaningful subgroups. Global parameters are shared across subjects.

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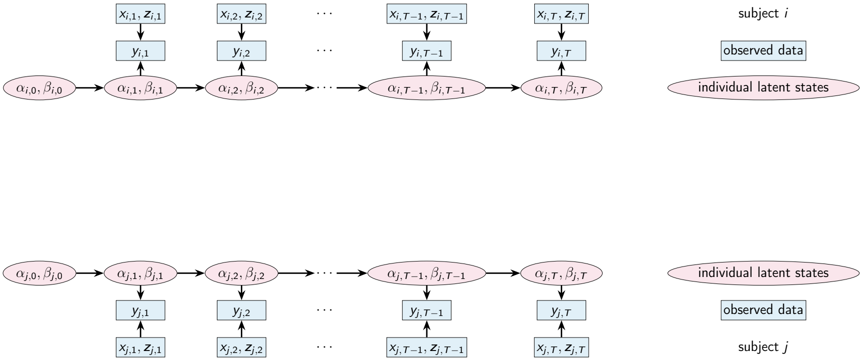

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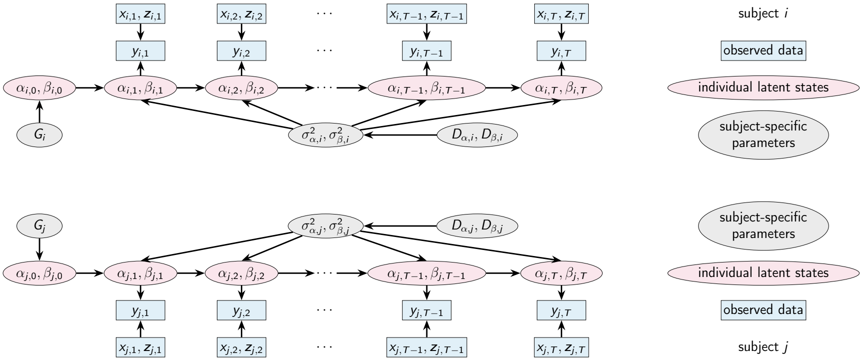

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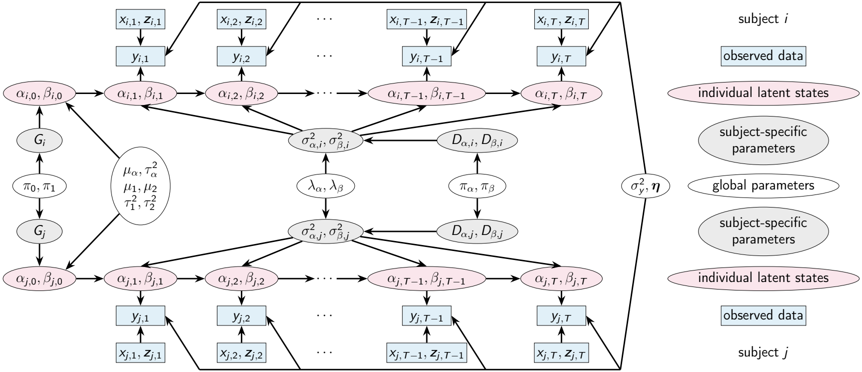

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Posterior Distribution

Target: the joint posterior over all unknowns given the data

$$p\Big(\{\boldsymbol{\theta}_{i,0:T}, G_i, D_{\alpha,i}, D_{\beta,i}, \sigma^2_{\alpha,i}, \sigma^2_{\beta,i}\}_{i=1}^{N}, \boldsymbol{\gamma}, \sigma^2_y, \boldsymbol{\eta} \,\Big|\, \text{data}\Big)$$

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Challenges:

- Discrete subgroup indicators $(G_i, D_{\alpha,i}, D_{\beta,i})$

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Challenges:

- Discrete subgroup indicators
- High-dimensional latent state trajectories ($2N(T+1)$ states)

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Solution:

Partially Collapsed Hybrid MCMC that combines reversible-jump, Metropolis–Hastings, forward filtering backward sampling (FFBS) and Gibbs.

MCMC Sampler: Overview

start → **Initialize MCMC at $m = 0$**

Set initial values for $\{G_i^{(0)}, D_{\alpha,i}^{(0)}, D_{\beta,i}^{(0)}, \sigma_{\alpha,i}^{2(0)}, \sigma_{\beta,i}^{2(0)}\}_{i=1}^{N}$, $\sigma_y^{2(0)}, \boldsymbol{\eta}^{(0)}, \boldsymbol{\gamma}^{(0)}$.

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↓ for iteration $m = 1, 2, \ldots$

For each individual $i = 1, \ldots, N$ within iteration m

(a) Reversible-jump update for baseline risk evolution $D_{\alpha,i}^{(m)}$ and $\sigma_{\alpha,i}^{2(m)}$

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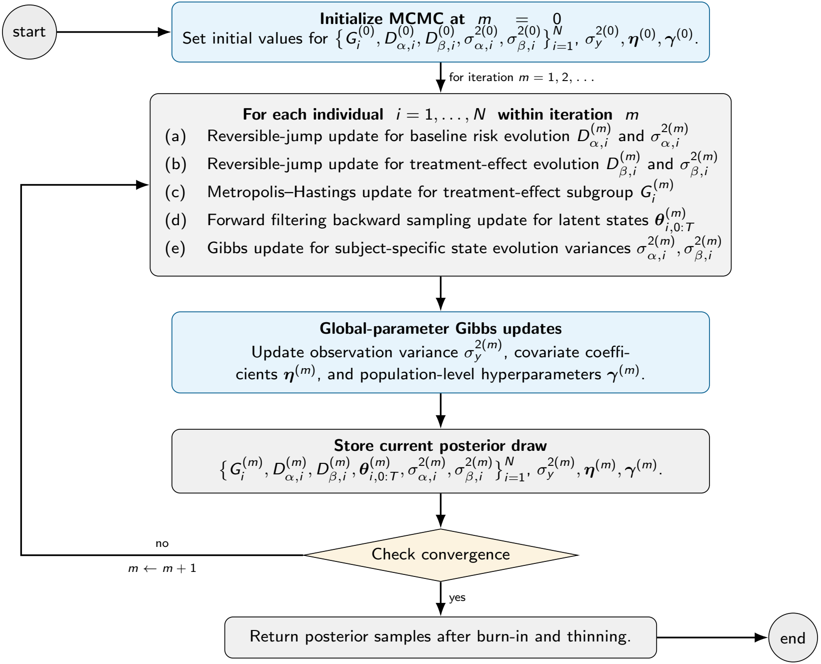
Check convergence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Check convergence
no
$m \leftarrow m + 1$
yes
Return posterior samples after burn-in and thinning.
end

Prediction for Individuals Under Follow-Up

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Key features:

- Predictive uncertainty grows with forecast horizon q
- **Clinically meaningful**: adapts to each patient's predictability

Online Learning for New Individuals

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Problem: a new patient enrolls after the model has been fitted to the original cohort

- Refitting the entire model from scratch is computationally expensive
- We want to *efficiently update* using the new individual's data

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- Need efficient updating

Sequential Monte Carlo (SMC) Approach

1. **Start from** MCMC posterior draws (equally weighted particles)
2. **Augment** each particle with new-subject-specific quantities: $(G_{i^\star}, D_{\alpha,i^\star}, D_{\beta,i^\star}, \sigma^2_{\alpha,i^\star}, \sigma^2_{\beta,i^\star})$
3. **Reweight** particles using new individual's likelihood
4. **Resample** if effective sample size is low
5. **Perturb** via short MCMC rejuvenation

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Sequential Monte Carlo (SMC) Approach

1. MCMC particles
2. **Augmentation**
3. **Reweighting**
4. **Resampling**
5. **Perturbation**

Output: posterior subgroup membership probabilities and predictive forecasts for the new patient — **without refitting** the full model.

Simulation Design

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Data-generating mechanism:

- $N = 120$ individuals, each observed at $T = 100$ time points
- $N_{\mathrm{new}} = 30$ additional out-of-sample individuals

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- $D_{\alpha,i}$: 80% dynamic; $D_{\beta,i}$: 60% dynamic (**non-stationarity**)

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Three evaluation **goals**:

1. **Inference**: parameter estimation and subgroup recovery

Simulation Design

Data-generating mechanism:

- $N = 120$, $T = 100$
- $N_{\mathrm{new}} = 30$
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3. **Online learning**: sequential updating for new individuals (vs. batch MCMC)

100 independent simulation runs for estimation; representative run for forecasting.

Parameter Estimation: Global Parameters

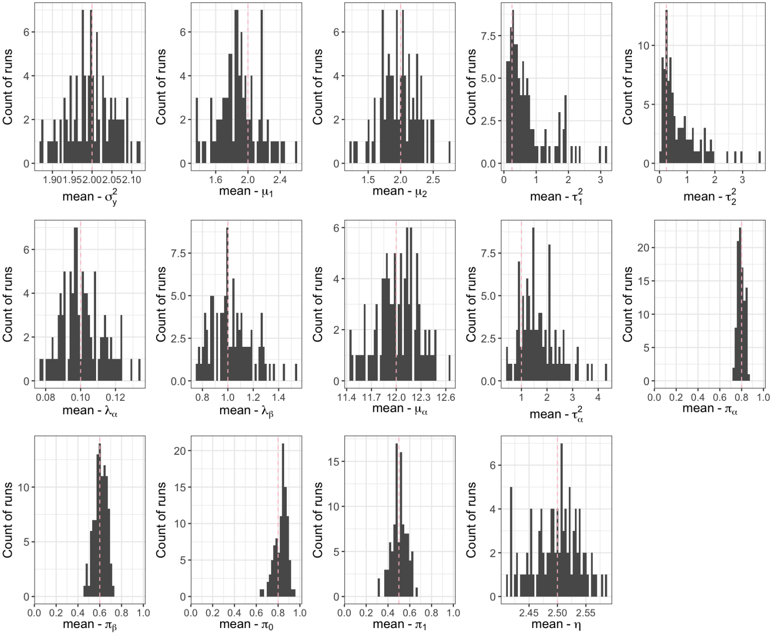

Figure 2: Run-level posterior means of global parameters across 100 independent runs.

Subgroup Classification: Posterior Probabilities

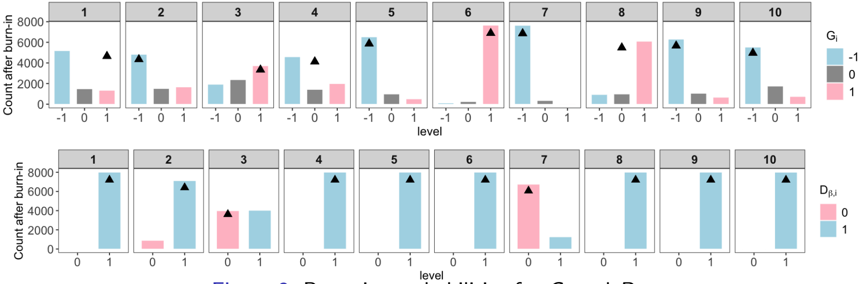

Figure 3: Posterior probabilities for G_i and $D_{\beta,i}$.

- Bar heights show posterior support for each possible subgroup value.
- Uncertain cases are visible through diffuse posterior probabilities, not hidden by a single hard label.

Takeaway: posterior probabilities quantify classification uncertainty.

Forecasting: Proposed Model vs. Bayesian GLMM

Table 2: Forecasting performance by step (within-subject holdout).

Step	Method	RMSE	MAE	Coverage	Avg. 95% Width
1	**HBM**	**3.05**	**2.40**	0.96	12.54
1	GLMM	15.59	10.98	0.85	41.14
3	**HBM**	**5.16**	**3.72**	0.97	18.41
3	GLMM	16.93	11.51	0.86	41.10
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- The proposed model has lower prediction error at each horizon.
- Predictive intervals are adaptive: wider for longer horizons, but not uniformly over-wide.

Sequential Updating: SMC vs. Batch MCMC

Subgroup classification:

- SMC closely reproduces batch posterior classifications
- Discrepancies occur when batch MCMC itself is uncertain
- **Computational efficiency:** 44.5% reduction in runtime

Table 3: SMC misclassification.

Subject	Indicator
123	$G_i, D_{\beta,i}$
125	$D_{\beta,i}$
126	G_i
129	$D_{\beta,i}$
130	$G_i, D_{\beta,i}$
132	$G_i, D_{\beta,i}$
138	$G_i, D_{\beta,i}$
139	$D_{\beta,i}$
140	G_i
144	$D_{\beta,i}$
146	G_i
147	G_i

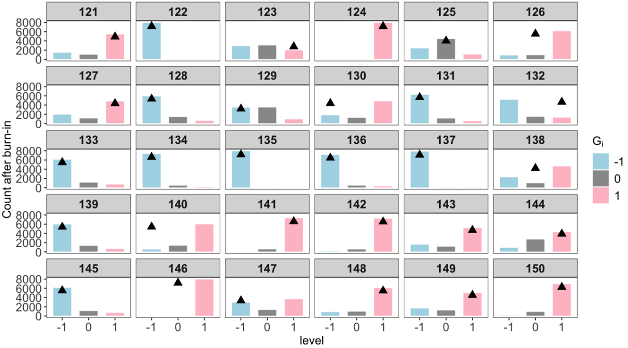

Figure 4: Batch MCMC posterior of G_i for new subjects. ▲ = true values.

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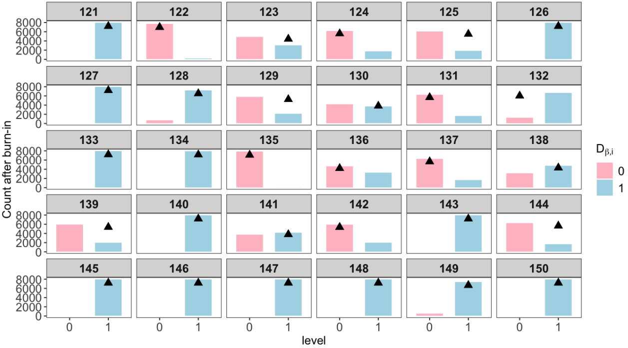

Figure 4: Batch MCMC posterior of $D_{\beta,i}$ for new subjects. ▲ = true values.

Summary of Contributions

1. **Hierarchical Bayesian state-space model** for *individualized*, *time-varying* treatment effects in mobile health data

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Summary of Contributions

1. **Hierarchical Bayesian state-space model** for *individualized*, *time-varying* treatment effects in mobile health data
2. **Clinically interpretable classification**
3. **Partially collapsed hybrid MCMC sampler** combining reversible-jump, Metropolis–Hastings, FFBS, and Gibbs steps

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1. **Hierarchical Bayesian state-space model** for *individualized*, *time-varying* treatment effects in mobile health data
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4. **Posterior predictive forecasting** with adaptive, patient-specific uncertainty

Summary of Contributions

1. **Hierarchical Bayesian state-space model** for *individualized*, *time-varying* treatment effects in mobile health data
2. **Clinically interpretable classification**
3. **Partially collapsed hybrid MCMC sampler**
4. **Posterior predictive forecasting**
5. **Sequential Monte Carlo updating** for new individuals without refitting the full model

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Broader Impact

This work contributes to **precision medicine** by supporting **individualized, dynamic intervention strategies** that adapt to patients' changing risk profiles over time.

Limitation and Future Directions

- **Non-Gaussian outcomes**: binary, count, or zero-inflated data commonly arising in psychiatric applications
- **Richer state evolution**: mean reversion, regime switching, or covariate-dependent dynamics
- **Robustness**: missing data and irregular observation times in EMA studies
- **Real-data application**: apply the proposed framework to the ARC study dataset

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Thank You!

Questions and comments are welcome.

Scan the QR code to visit my personal website and download these slides.
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